EXPERIMENT 5

CO1 LAB-5

Program:

```
from collections import deque
def solve():
    q = deque([(3, 3, 1, [])])
    visited = set()
    while q:
        m, c, b, path = q.popleft()
        if (m, c, b) in visited:
            continue
        visited.add((m, c, b))
        path = path + [(m, c, b)]
        if (m, c, b) == (0, 0, 0):
            print(path)
            return
        for x, y in [(1,0), (2,0), (0,1), (0,2), (1,1)]:
            nm = m - x if b else m + x
            nc = c - y if b else c + y
            nb = 1 - b
            if 0 <= nm <= 3 and 0 <= nc <= 3:
                if (nm == 0 or nm >= nc) and (3-nm == 0 or 3-nm >= 3-nc):
                    q.append((nm, nc, nb, path))

solve()
```

Output:

```
[(3, 3, 1), (3, 1, 0), (3, 2, 1), (3, 0, 0), (3, 1, 1), (1, 1, 0), (2, 2, 1), (0, 2, 0), (0, 3, 1), (0, 1, 0), (1, 1, 1), (0, 0, 0)]
```